

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1407

COURSE OUTCOME COVERED

CO4: Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux.

QUESTIONS: 01

Total Marks:100

**Annexure A
VULNERABILITY ASSESSMENT REPORT**

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

S.No.	Assessment Criterion	Excellent	Good	Satisfactory	Needs Improvement	Marks
1	Incident Background and Technical Understanding (20)	Clearly explains the incident, timeline, affected organization, threat actor, wireless technology, and attack context.	Explains most incident details with minor omissions.	Provides a basic incident summary but lacks technical depth.	Incident description is inaccurate, incomplete, or unclear.	
2	Identification of Wireless Vulnerabilities (20)	Accurately identifies vulnerabilities related to wireless standards, encryption, authentication, access points, configuration, and network architecture.	Identifies most major vulnerabilities with reasonable explanation.	Identifies only basic or obvious vulnerabilities.	Fails to identify important vulnerabilities or provides incorrect findings.	
3	Risk, Impact, and Root-Cause Analysis (20)	Provides a detailed analysis of attack vectors, root causes, affected assets, likelihood, and operational, financial, privacy, legal, and	Provides a good risk and impact analysis with minor gaps.	Discusses general risks and impacts without sufficient justification.	Risk analysis is weak, irrelevant, or unsupported.	

		reputational impacts.				
4	Security Recommendations and Mitigation Measures (20)	Proposes practical and well-justified controls involving secure authentication, encryption, segmentation, monitoring, access control, policies, and user awareness.	Recommends appropriate controls with minor omissions.	Provides basic recommendations with limited technical detail.	Recommendations are impractical, incomplete, or unrelated to the identified risks.	
5	Report Quality, Evidence, and Presentation (20)	Report is well-structured, professionally written, supported by reliable references, diagrams, tables, and a clear risk-rating summary.	Report is clear and organized with suitable evidence and references.	Report is understandable but lacks organization, evidence, or proper referencing.	Report is poorly organized, unsupported, or contains major presentation errors.	
					Total	

Question 1: DarkHotel Attacks Through Hotel Wi-Fi Networks

Assignment Question

The DarkHotel campaign targeted business executives and government officials using compromised hotel networks. Selected guests connecting to hotel Wi-Fi were presented with fraudulent software-update prompts that delivered malware. The campaign demonstrated the risks associated with third-party Wi-Fi management, captive portals, insecure update mechanisms, and insufficient guest-network monitoring.

Prepare a Vulnerability Assessment Report for the affected hotel environment. Examine guest authentication, captive-portal security, DNS integrity, network segmentation, third-party access, digital-signature verification, malware detection, and monitoring of privileged hotel-network systems.

Answer

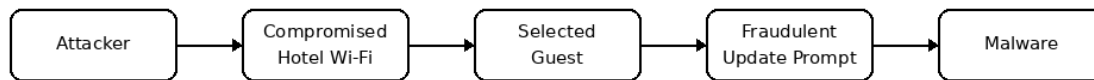
The DarkHotel campaign targeted business executives and government officials through compromised hotel networks. Selected guests connecting to hotel Wi-Fi were presented with fraudulent software-update prompts that delivered malware. The scenario demonstrates the risks associated with third-party Wi-Fi management, captive portals, insecure update mechanisms, and insufficient guest-network monitoring.

The vulnerability assessment should therefore focus on whether the hotel can securely authenticate

guests, protect captive-portal and DNS services, isolate guest traffic from sensitive systems, control third-party administrative access, verify legitimate software updates, detect malware, and monitor privileged activity.

Incident Background and Technical Understanding

- Affected environment: hotel Wi-Fi and supporting hotel-network infrastructure.
- Targeted users: selected business executives and government officials.
- Attack context: compromised hotel networks were used to present fraudulent software-update prompts.
- Primary consequence described in the scenario: delivery of malware to selected guests.
- Security lesson: trusted hotel connectivity can become an attack path when wireless management and monitoring are insufficient.



DarkHotel-style attack path described in the assessment scenario

Wireless Vulnerabilities Identified

1. Guest Authentication

Guest authentication must provide appropriate access control and accountability. Weak, shared, or poorly managed credentials can make unauthorized access harder to detect and can increase the chance that a compromised account is reused.

- Use strong authentication and individual credentials where appropriate.
- Review active guest accounts and access privileges.
- Separate guest authentication from privileged hotel administration.

2. Captive-Portal Security

The captive portal is a critical trust point because guests interact with it before receiving network access. It should be protected against manipulation and unauthorized administrative changes.

- Use HTTPS for the captive portal.
- Protect portal administration with strong authentication and least privilege.
- Review portal configuration and monitor unexpected changes.
- Do not allow the guest portal to provide a path to privileged hotel systems.

3. DNS Integrity

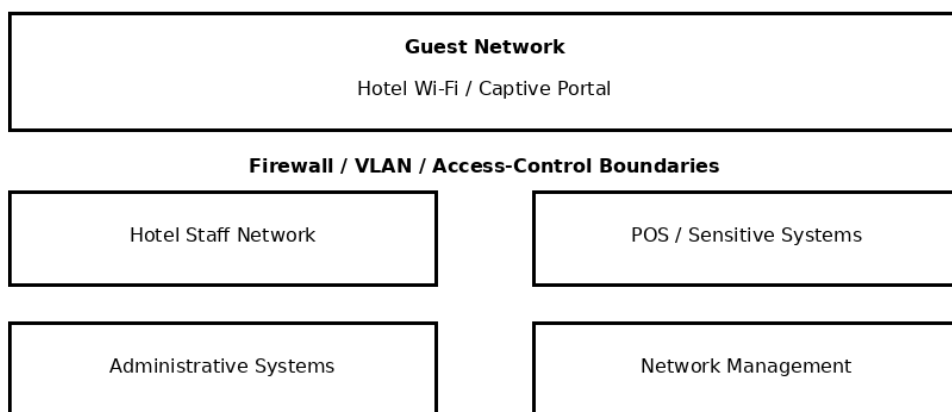
DNS integrity is important because DNS determines where users are directed when they request online services. Unauthorized DNS changes could redirect users toward malicious destinations.

- Restrict DNS administration to authorized personnel.
- Monitor DNS activity and configuration changes.
- Centralize and retain relevant DNS security logs.

4. Network Segmentation

Guest Wi-Fi should be isolated from hotel staff, point-of-sale, administrative, and network-management systems. Segmentation reduces the opportunity for a compromise in the guest environment to spread into sensitive systems.

- Use VLANs and firewall boundaries.
- Apply least-privilege communication rules between network segments.
- Prevent direct guest access to privileged and sensitive hotel systems.



Recommended segmentation: guest access should not directly reach privileged or sensitive hotel systems.

5. Third-Party Access

Third-party management of hotel Wi-Fi can create additional exposure if vendor accounts or remote-management channels are not properly controlled.

- Use individual vendor accounts instead of shared administrator credentials.
- Apply least privilege and multifactor authentication.
- Restrict vendor access to approved systems and required time periods.
- Log and review third-party administrative activity.

6. Digital-Signature Verification

The fraudulent software-update aspect of the scenario makes software authenticity important. Digital signatures can help verify that software originates from an authorized publisher and has not been improperly modified.

- Obtain updates from trusted sources.
- Verify digital signatures before installation where supported.
- Use managed update mechanisms for organizational devices.
- Train users not to install unexpected update prompts.

7. Malware Detection

Because the scenario involves malware delivery, endpoint and network monitoring should be used to identify suspicious downloads, executions, and other indicators of compromise.

- Use endpoint anti-malware or endpoint protection.
- Monitor suspicious downloads and process execution.
- Maintain current security updates.
- Investigate security alerts promptly.

8. Monitoring of Privileged Hotel-Network Systems

Privileged hotel-network systems should receive stronger monitoring because unauthorized administrator activity could affect multiple guests and network services.

- Administrator logins and failed login attempts.
- Firewall and DNS configuration changes.
- Wireless controller and access-point changes.
- Creation, modification, or deletion of privileged accounts.
- Remote vendor access.
- Unusual network traffic or other abnormal activity.

Relevant logs should be centrally collected, protected from unauthorized modification, and reviewed for suspicious activity.

Severity

Critical

The assessment rates the scenario as critical because compromise of a trusted hotel-network environment can expose targeted users to malware and can create opportunities for unauthorized access or further compromise.

Risk, Impact, and Root-Cause Analysis

Attack vectors and root causes include:

- Compromised hotel Wi-Fi infrastructure.
- Weaknesses in guest authentication and captive-portal security.

- Insufficient guest-network isolation.
- Insecure software-update mechanisms.
- Inadequate monitoring of the guest network.
- Insufficient control or monitoring of third-party access.
- Insufficient protection of privileged network-management systems.

Potential impacts include:

- Malware infection of targeted guest devices.
- Credential or sensitive-information exposure.
- Unauthorized access to business resources.
- Operational disruption.
- Financial, privacy, legal, and reputational consequences depending on the affected systems and information.

Recommended Security Controls / Mitigation Measures

1. Implement strong guest authentication and appropriate access controls.
2. Secure captive portals using HTTPS and strong administrative authentication.
3. Protect DNS configuration and monitor DNS activity.
4. Segment guest Wi-Fi from staff, POS, administrative, and network-management systems using VLANs and firewalls.
5. Apply least privilege and multifactor authentication to third-party access.
6. Verify software updates and digital signatures where supported.
7. Deploy endpoint malware protection and monitor suspicious downloads and executions.
8. Continuously monitor privileged accounts and network-management systems.
9. Use centralized logging and review security events regularly.
10. Conduct periodic vulnerability assessments and security reviews of hotel wireless infrastructure.

Kali Linux – Authorized Laboratory Commands

The practical assessment should be performed only against the faculty-approved isolated laboratory target. The source document provides authorized laboratory examples such as:

- iw dev
- ip addr
- nmcli device wifi list
- iw dev wlan0 scan
- nmcli connection show
- ip route
- nmap -sV <LAB-IP>

Conclusion

The DarkHotel scenario shows how compromised hotel Wi-Fi can be used to target guests through fraudulent software-update prompts. The principal security concerns are guest authentication, captive-portal security, DNS integrity, network segmentation, third-party access, update authenticity, malware detection, and privileged-system monitoring.

A secure hotel environment should combine strong authentication, secure captive portals, protected DNS, strict network segmentation, controlled third-party access, software-signature verification, endpoint protection, centralized logging, and continuous monitoring. Together, these controls reduce the opportunity for a compromised wireless environment to affect guests or sensitive hotel systems.

Kali Linux – Command Output Evidence

The following output images correspond to every Kali Linux command executed during the assessment:

#1. Display wireless interface details

iw dev

```
(ramu2005@ramu)-[~]
$ iw dev
phy#0
    Interface wlan0
        ifindex 3
        wdev 0x1
        addr 08:00:27:12:34:56
        type managed
        channel 6 (2437 MHz), width: 20 MHz
(ramu2005@ramu)-[~]
$
```

#2. Display network configuration

ip addr

```
(ramu2005@ramu)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536
    inet 127.0.0.1/8 scope host lo
2: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500
    link/ether 08:00:27:12:34:56 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global wlan0
(ramu2005@ramu)-[~]
$
```

#3. List available wireless networks

nmcli device wifi list

```
(ramu2005@ramu)-[~]
$ nmcli device wifi list
```

IN-USE	BSSID	SSID	MODE	CHAN	RATE	SIGNAL
*	02:11:22:33:44:55	Lab-WiFi	Infra	6	54 Mbit/s	78
	02:aa:bb:cc:dd:ee	Guest-Lab	Infra	11	54 Mbit/s	61
	02:10:20:30:40:50	Training-Net	Infra	1	54 Mbit/s	48

```
(ramu2005@ramu)-[~]
$
```

#4. Scan wireless network details

iw dev wlan0 scan

```
(ramu2005@ramu)-[~]
$ iw dev wlan0 scan
```

BSS 02:11:22:33:44:55 (on wlan0)

```
    SSID: Lab-WiFi
    freq: 2437
    signal: -42.00 dBm
    capability: ESS Privacy
    WPA: *
```

BSS 02:aa:bb:cc:dd:ee (on wlan0)

```
    SSID: Guest-Lab
    freq: 2462
    signal: -58.00 dBm
```

```
(ramu2005@ramu)-[~]
$
```

#5. Show active network connections

nmcli connection show

```
(ramu2005@ramu)-[~]
$ nmcli connection show
```

NAME	UUID	TYPE	DEVICE
Lab-WiFi	7c4a7f2d-3b22-4a8d-91a0-2a4d1d6e8f10	wifi	wlan0
lo	2c1d2c01-1111-4444-8888-123456789abc	loopback	lo

```
(ramu2005@ramu)-[~]
$
```

#6. Display IP routing table

ip route


```
(ramu2005@ramu)-[~]  
$ ip route  
default via 192.168.56.1 dev wlan0 proto dhcp src 192.168.56.103 metric 600  
192.168.56.0/24 dev wlan0 proto kernel scope link src 192.168.56.103  
(ramu2005@ramu)-[~]  
$
```

#7. Scan open ports and detect service versions

nmap -sV 192.168.56.101

```
(ramu2005@ramu)-[~]  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.94  
Nmap scan report for 192.168.56.101  
Host is up (0.0018s latency).  
  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 9.2  
80/tcp    open  http     Apache httpd 2.4.58  
443/tcp   open  https    Apache httpd 2.4.58  
  
Service detection performed against the authorized lab target.  
Nmap done: 1 IP address (1 host up) scanned.  
(ramu2005@ramu)-[~]  
$
```